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Where Has All the Data Gone?

Contents

1 Big picture

- **Question.** How can we measure the quantity or precision of financial data that investors process about different stocks, and how has that data changed across firms over time?
- **Main empirical fact.** Data processing has diverged. Investors appear to process much more data about large high-growth firms over time, while data about other types of firms has stagnated or grown much less.
- **Why a structural measure is needed.** Price informativeness is not the same as data. A price can be informative because investors process more data, but also because the firm's cash-flow growth, cash-flow volatility, or price volatility changes.
- **Core tool.** The paper builds a noisy rational expectations model that maps a reduced-form price informativeness moment into three components: volatility, growth, and data.
- **Key decomposition.** Stock-specific price informativeness is written as

$$\mathrm{PINF}_{js} = \underbrace{\frac{\Sigma_{jd}}{\mathrm{StdDev}(P_{fj1})}}_{\text{volatility}} \underbrace{\frac{g_j^s}{r - g_j}}_{\text{growth}} \underbrace{\left(1 - \frac{\Sigma_j}{\Sigma_{jd}}\right)}_{\text{data}}.$$

- **Data measure.** The data component is the fraction of prior cash-flow uncertainty removed by investors' information. A value near zero means investors learn little beyond priors; a value near one means investors learn a lot.
- **Main result.** For large high-growth firms, the data component rises sharply from the 1960s to the 2010s. For small firms and low-growth firms, the data component rises much less or remains low.
- **Mechanism.** The value of data rises with firm size and growth. Large high-growth firms become especially worth learning about because investors can take large positions and because growth magnifies the price impact of cash-flow information.
- **Broader implication.** The financial data revolution may not benefit all firms equally. More data can make prices more informative for firms that are already large and high-growth, while leaving smaller or less glamorous firms relatively neglected.

2 Incremental contribution of the paper

- **From price informativeness to data precision.** Prior empirical work often measures whether prices forecast future cash flows. This paper shows that such measures confound true investor data with growth, volatility, and price scaling effects.
- **A structural measurement formula.** The paper derives a tractable formula that backs out the amount of data processed from observable price and cash-flow moments. This turns a reduced-form covariance into a model-consistent data measure.
- **A new fact: data divergence.** The paper documents that data processing increasingly concentrates on large high-growth firms. This is not simply a rise in informativeness for all firms; it is a cross-sectional divergence.
- **Separating price information and private data.** The paper decomposes total data into information learned from prices and net private data. It shows that the rise for large high-growth firms is driven mainly by net private information.
- **A value-of-data explanation.** The paper does not stop at measurement. It derives a model-implied value of information and shows that data became more valuable for large high-growth firms because large firms became even larger and growth amplified the payoff to learning.

- **Clarifying other proxies.** The paper explains why analyst coverage, comovement, synchronicity, and price informativeness are useful but incomplete proxies. These measures often point in the same broad direction, but the structural measure gives the magnitude in units of data precision.
- **Macro-finance relevance.** The paper highlights a new dimension of financial-market inequality: some assets receive more data, analyst effort, and price discovery, potentially changing capital allocation and real investment efficiency across firms.

3 Data and measurement

3.1 Sample and sources

The sample covers U.S. publicly traded firms from 1962 to 2016. Stock prices come from CRSP and accounting variables come from Compustat. Financial firms are excluded. Prices are measured at the end of March and accounting variables at the end of the previous fiscal year so that accounting information used as controls is plausibly available to market participants.

- Equity valuation is measured as market capitalization divided by total assets.
- Cash flows are proxied by EBIT divided by total assets.
- Both ratios are winsorized at 1%.
- Cash-flow variables are deflated to remove inflation-driven predictability.
- Common market components are removed by projecting firm-level cash flows and prices on analogous S&P 500 market variables and using residuals.

Interpretation: The empirical object is stock-specific information. Removing aggregate components is important because the paper wants to measure data about firm-specific cash flows, not data about the market as a whole.

3.2 Sorting firms by size and growth

Firms are sorted into four groups by size and growth:

- **Large versus small:** large firms are the 500 largest firms by market capitalization; the remaining firms are small.
- **High-growth versus low-growth:** high-growth firms are in the bottom 30% of book-to-market; low-growth firms are in the top 30% of book-to-market.
- The four groups are small/high-growth, large/high-growth, small/low-growth, and large/low-growth.

Interpretation: Size and growth are not arbitrary sorts. In the model, size raises the scale of positions that informed investors can take, while growth raises the sensitivity of value to cash-flow news. These are precisely the dimensions along which data should be more valuable.

3.3 Measuring stock-specific price informativeness

For each group j and decade, the paper estimates a regression of future cash flows on prices and controls:

$$\frac{E_{f,j,t+s}}{A_{f,j,t}} = \alpha_j + \beta_{j,s}\frac{M_{f,j,t}}{A_{f,j,t}} + \gamma_j X_{f,j,t} + \varepsilon_{f,j,t+s}. \tag{1}$$

The paper uses $s = 3$. The stock-specific price informativeness measure is:

$$\mathrm{PINF}_{js} = \beta_{j,s}\sigma_j^{M/A}, \tag{2}$$

where $\sigma_j^{M/A}$ is the cross-sectional standard deviation of market capitalization scaled by assets, conditional on controls.

Interpretation: This moment captures how much stock prices forecast future cash flows after conditioning on public controls. The model then tells us how to convert this moment into data precision.

3.4 Cash-flow persistence, growth, and volatility

The cash-flow process is estimated by group and decade. The model writes current cash flow as:

$$d^*_{fj1} = g_j d^*_{fj0} + \bar{\varepsilon}_{fj1} + \varepsilon_{fj1}, \qquad \varepsilon_{fj1} \sim N(0, \Sigma_{jd}). \tag{3}$$

The parameter g_j captures persistence or growth. The variance Σ_{jd} captures the variance of the stock-specific cash-flow innovation.

Interpretation: Growth and volatility matter because they scale how a given amount of data affects price informativeness. Failing to adjust for them would confuse changes in data with changes in firm fundamentals.

3.5 Data component

The paper defines the data component as:

$$\mathrm{Data}_j = 1 - \frac{\Sigma_j}{\Sigma_{jd}}, \tag{4}$$

where Σ_{jd} is prior variance and Σ_j is posterior variance after investors process data.

Interpretation: This is a normalized precision measure. It is zero if investors learn nothing about firm-specific cash-flow innovations and approaches one if investors learn almost everything.

3.6 Price information versus private data

The total data measure includes both information extracted from prices and net private information. The paper separates them using:

$$K_j = \bar{\Sigma}_j^{-1} - \Sigma_{jd}^{-1} - \Sigma_{jp}^{-1}, \tag{5}$$

where Σ_{jp}^{-1} is the information contained in the price signal and K_j is net private data.

Interpretation: This decomposition asks whether the data revolution came from learning through prices or from investors processing private data outside prices. The main result is that large high-growth firms' rise comes mostly from net private information.

4 Structural model and empirical specifications

4.1 Noisy rational expectations environment

A unit mass of investors trades multiple stocks. Each stock is a claim to a dividend stream. Supply is random, investors have mean-variance preferences, and investors receive data with precision that may differ across assets and investors.

The value of a stock after the first-period dividend is:

$$V_{fj1}^{*} = \frac{r}{r - g_j} d_{fj1}^{*}. \tag{6}$$

Interpretation: The Gordon-growth term $r/(r - g_j)$ is central. It explains why the same cash-flow information can matter more for high-growth firms: it moves valuation more strongly.

4.2 Equilibrium price

The equilibrium price is linear in the firm-specific cash-flow innovation and supply noise:

$$rP_{fj1}^{*} = A_{fj} + B_j \varepsilon_{fj1} + C_j \tilde{x}_{fj}. \tag{7}$$

The sensitivity to cash-flow news is:

$$B_j = \frac{r}{r - g_j} \left(1 - \frac{\Sigma_j}{\Sigma_{jd}}\right). \tag{8}$$

Interpretation: The price loads on cash-flow news only if investors process data. If posterior variance equals prior variance, the data term is zero and price cannot reflect cash-flow innovations. If investors are perfectly informed, the loading approaches the Gordon-growth multiple.

4.3 Price informativeness decomposition

The model implies:

$$\text{PINF}_{js} = \frac{\Sigma_{jd}}{\text{StdDev}(P_{fj1})} \frac{g_j^s}{r - g_j} \left(1 - \frac{\Sigma_j}{\Sigma_{jd}}\right). \tag{9}$$

The three components are:

- **Volatility:** $\Sigma_{jd}/\text{StdDev}(P_{fj1})$.
- **Growth:** $g_j^s/(r - g_j)$.
- **Data:** $1 - \Sigma_j/\Sigma_{jd}$.

Interpretation: A rise in PINF does not automatically mean more data. It might instead reflect a change in growth or volatility. The structural decomposition is designed to isolate the data term.

4.4 Value of information

The paper derives the initial value of information, VI_j, from the model. In compact form, the value rises with the expected profitability of informed trading and the scale of tradable positions:

$$VI_j \propto \frac{1}{2} \left(\frac{r}{r - g_j}\right)^2 \bar{x}_j \times \text{cash-flow uncertainty and learning value.} \tag{10}$$

Interpretation: Data are worth more when firms are larger, more volatile, or higher growth. Large high-growth firms combine a large tradable scale with a high valuation sensitivity to cash-flow information.

5 Identification and empirical/theoretical design

This is not a causal treatment-effect paper. The identification problem is a measurement problem: how can the researcher infer the precision of investors' data from observed prices and cash flows? The paper solves this by imposing a simple equilibrium model that links data precision to observable moments.

5.1 What identifies data precision

Data precision is identified from the combination of three observed or estimated objects:

1. stock-specific price informativeness, PINF_{js};
2. the growth component, $g_j^s/(r-g_j)$;
3. the volatility component, $\Sigma_{jd}/\text{StdDev}(P_{fj1})$.

Rearranging the model equation gives:

$$\text{Data}_j = 1 - \frac{\Sigma_j}{\Sigma_{jd}} = \frac{\text{PINF}_{js}}{(\Sigma_{jd}/\text{StdDev}(P_{fj1}))\left(g_j^s/(r-g_j)\right)}. \tag{11}$$

Interpretation: The identifying variation is not an instrument or policy shock. It is the structural mapping between observed price-cash-flow predictability and the latent amount of data investors process.

5.2 Why the decomposition matters

A reduced-form regression coefficient can rise because investors process more data, because growth amplifies cash-flow news, or because price volatility changes. The paper's decomposition allows the researcher to say which force is responsible.

Interpretation: This is why the main empirical fact is stronger than simply saying price informativeness increased. The paper shows that large high-growth firms experienced a true rise in inferred data, even though changes in growth and volatility often work against the trend.

5.3 Assumptions behind the measurement

The measurement rests on several assumptions:

- prices are generated by a noisy rational expectations equilibrium;
- investors trade on data about firm-specific cash-flow innovations;
- the model's group-level parameters are meaningful within size-growth groups;
- the common market component can be removed using S&P 500 aggregate variables;
- future cash-flow regressions recover the relevant price informativeness moment;
- the riskless rate and cash-flow persistence estimates are reasonable inputs for the structural mapping.

Interpretation: These assumptions do not prove that the measure is the only possible measure of data. They make the measure interpretable in terms of a standard equilibrium model of information acquisition.

5.4 Validation and robustness

The paper validates the measurement in several ways:

- **Component check:** Table 2 and Figure 1 show that the large high-growth divergence comes from the data component, not from growth or volatility.
- **Source check:** Table 3 and Figure 2 show that the increase comes mainly from net private data rather than price information.
- **Proxy check:** Figure D1 shows that analyst coverage rises more for high-growth firms, especially within large firms.
- **Robustness check:** Figure B1 shows that results are similar when the interest rate varies over time rather than being fixed.
- **Size check:** Figure E3 and Figure E4 support the interpretation that size, not only S&P 500 index inclusion, matters for informativeness.

Interpretation: The validation exercises do not turn the paper into a causal identification design. Instead, they make the structural measurement credible by showing that multiple related facts point in the same direction.

5.5 Main threat to interpretation

The biggest threat is that model misspecification may attribute changes in prices or cash flows to data when they reflect other forces such as changing risk premia, market power, accounting changes, or unmodeled heterogeneity.

Interpretation: The paper addresses this by removing common components, using controls in the PINF regressions, comparing to alternative proxies, and presenting robustness exercises. Still, the estimates should be interpreted as model-implied data precision, not a direct observation of investor datasets.

6 Tables and figures

6.1 Table 1: Estimated cash flow parameters; Table 2: Stock-specific price informativeness and its components; Figure 1: Data divergence

Read:

- Table 1 reports cash-flow persistence/growth and innovation variances by decade and size-growth group.
- Large high-growth firms have very high persistence early in the sample, but their growth component declines over time.
- Table 2 reports PINF, volatility, growth, and the inferred data component.
- The data component for large high-growth firms rises from 0.36 in the 1960s to 0.95 in the 2010s.
- Small and low-growth groups do not show a comparable increase in data.
- Figure 1 makes the contrast visually clear: large high-growth firms have the steepest data trend, while their growth trend actually falls.

Economic message: The central finding is that the rise in informativeness for large high-growth firms is not mechanically caused by growth or volatility. The inferred data component rises sharply even when other components move in the opposite direction.

Interpretation: Table 1 is the input table for the structural correction. Table 2 and Figure 1 are the main measurement outputs. Together, they show why a reduced-form PINF trend is not enough: the same PINF number must be adjusted for growth and volatility before it can be read as data.

How it fits the paper: This is the paper's main empirical contribution. The financial data revolution is not uniform; it is concentrated in a narrow group of large high-growth firms.

Table 1
Estimated cash flow parameters: Persistence/growth g_j, variance of innovation Σ_{jd}

	1960s	1970s	1980s	1990s	2000s	2010s
Persistence g_j:						
Small/high growth	0.830	0.877	0.702	0.725	0.740	0.741
Large/high growth	0.988	0.981	0.954	0.949	0.917	0.912
Small/ low growth	0.829	0.697	0.538	0.572	0.669	0.636
Large/ low growth	0.901	0.865	0.853	0.813	0.828	0.851
Variance of Innovations Σ_{jd}:						
Small/high growth	0.005	0.007	0.019	0.022	0.017	0.013
Large/high growth	0.002	0.002	0.002	0.003	0.003	0.003
Small/ low growth	0.002	0.004	0.009	0.008	0.009	0.006
Large/ low growth	0.002	0.001	0.001	0.001	0.002	0.001

Persistence g_j is estimated by running regressions of cash flows on their lagged values, as specified in Equation (1). Σ_{jd} is estimated as the variance of residuals from a projection of cash flows on controls.

(a) Table 1: Estimated cash flow parameters: persistence/growth g_j, variance of innovation Σ_{jd}.

Table 2
Stock-specific price informativeness and its components

	1960s	1970s	1980s	1990s	2000s	2010s
PINF						
Small/high growth	0.012	0.003	0.003	0.008	0.013	0.007
Large/high growth	0.015	0.014	0.018	0.011	0.037	0.023
Small/ low growth	0.003	0.007	0.003	0.002	0.011	0.004
Large/ low growth	0.003	0.006	0.002	0.003	0.014	0.001
Volatility						
Small/high growth	0.005	0.008	0.012	0.013	0.011	0.010
Large/high growth	0.002	0.002	0.003	0.002	0.002	0.003
Small/ low growth	0.016	0.054	0.064	0.044	0.037	0.034
Large/ low growth	0.012	0.017	0.012	0.011	0.013	0.008
Growth						
Small/high growth	2.94	4.56	1.07	1.27	1.42	1.44
Large/high growth	25.87	21.34	12.33	11.31	7.17	6.69
Small/ low growth	2.91	1.03	0.32	0.41	0.84	0.66
Large/ low growth	5.91	4.06	3.60	2.54	2.88	3.53
Data						
Small/high growth	0.84	0.08	0.22	0.50	0.80	0.50
Large/high growth	0.36	0.40	0.57	0.44	0.95	0.95
Small/ low growth	0.06	0.12	0.12	0.10	0.36	0.17
Large/ low growth	0.04	0.09	0.04	0.10	0.38	0.03

The table reports structurally estimated values of the various terms in Equation (13) using cash flow parameters in Table 1. The left-hand side is estimated using Equations (14) and (15).

(b) Table 2: Stock-specific price informativeness and its components.

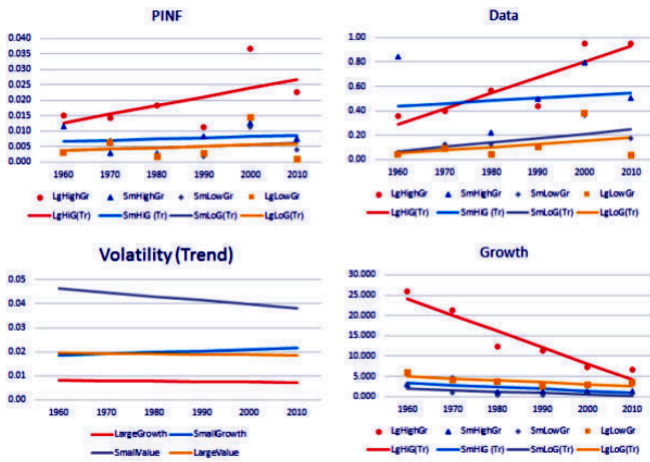


Figure 1: Data divergence: Trends in data and other components of stock-specific price informativeness. The figure visualizes Table 2 and shows that data rises most for large high-growth firms.

6.2 Table 3: Sources of information; Figure 2: Rising large/high-growth firm data come from net private information

Read:

- Table 3 decomposes total information into price information and net private data.
- Price information declines or remains small across many groups.
- Net private data rises sharply for large high-growth firms, reaching very large values by the 2010s.
- Figure 2 shows the fitted trends: the total information rise for large high-growth firms is almost entirely due to the net private information component.

Economic message: The data revolution in the paper is not simply that prices themselves reveal more information. It is that investors appear to process more private or non-price data about large high-growth firms.

Interpretation: This distinction matters because price information can be a public signal that everyone can use, while net private information reflects investor research, proprietary data, analyst effort, and other information-processing activity outside the price signal.

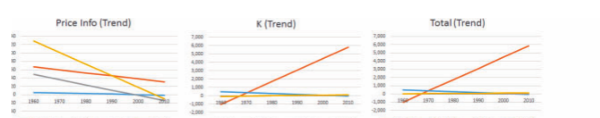
How it fits the paper: The table and figure support the paper's claim that financial markets have produced more data about some firms, not merely rearranged how information is revealed through prices.

Sources of information

	1960s	1970s	1980s	1990s	2000s	2010s
Price information, Σ_{jp}^{-1}:						
Small/high growth	10	1	0	1	1	0
Large/high growth	80	18	82	41	55	16
Small/ low growth	77	20	3	0	3	0
Large/ low growth	130	105	72	41	8	1
Net private information, K_j:						
Small/high growth	1013	12	15	45	236	75
Large/high growth	269	362	451	193	7151	5868
Small/ low growth	−49	13	14	13	59	33
Large/ low growth	−106	−33	−40	45	258	30

Price information and net private data are estimated using Equations (17) and (18), respectively.

(a) Table 3: Sources of information.



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ng large/high-growth firm data come from net private information
hical representation of the trends in the estimates reported in Table 3. For each component, the plot shows near trend fitted to the estimates from the table. Total is the sum of Price Information (Σ_{ip}^{-1}) and Net Private

(b) Figure 2: Rising large/high-growth firm data come from net private information.

6.3 Figure 3: Price informativeness by firm size and growth

Read:

- Figure 3 plots the reduced-form price informativeness measure from Bai, Philippon, and Savov style regressions.
- In the left panel, large high-growth firms become more informative over time, while large low-growth firms do not.
- In the right panel, small high-growth firms appear to decline in price informativeness, while small low-growth firms are relatively stable.
- The confidence intervals show that the cross-sectional pattern is noisy but directionally consistent with the structural data divergence result.

Economic message: Reduced-form price informativeness already suggests divergence, but it cannot explain whether the divergence comes from true data, growth, volatility, or market-pricing changes.

Interpretation: Figure 3 is a useful bridge to prior literature. It shows that the paper's new structural data measure is connected to existing price informativeness concepts, but the structural measure is more interpretable because it corrects for firm characteristics.

How it fits the paper: This figure helps convince readers that the structural result is not an artifact of the model alone. A familiar reduced-form measure points in a related direction.

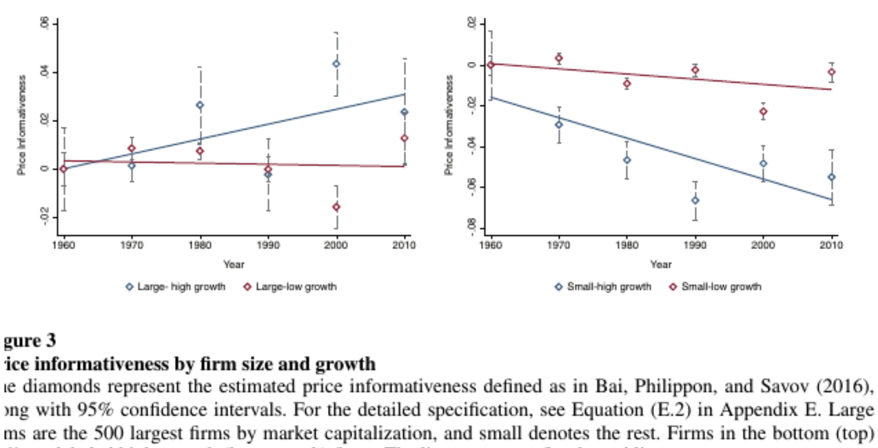


Figure 3: Price informativeness by firm size and growth. The figure plots reduced-form price informativeness trends by size-growth group.

6.4 Figure 4: The initial value of information, by asset class, over time

Read:

- Figure 4 plots the model-implied value of information, VI_j, by size-growth group and decade.
- The value of information for large high-growth firms rises sharply in the 1990s and 2000s.
- Large low-growth firms also become valuable to learn about when their size rises substantially.
- The small-firm lines remain close to zero, indicating that learning about small firms is much less valuable in the model.

Economic message: The figure provides an explanation for data divergence: investors learn more about assets for which information is most valuable. Large firms offer more scale, and high-growth firms amplify cash-flow news through valuation multiples.

Interpretation: Figure 4 turns the paper from a measurement exercise into a theory of why the measured patterns emerged. Data go where data are most valuable.

How it fits the paper: This figure is central to the incremental contribution: the paper both measures data and links its allocation to the economic value of learning.

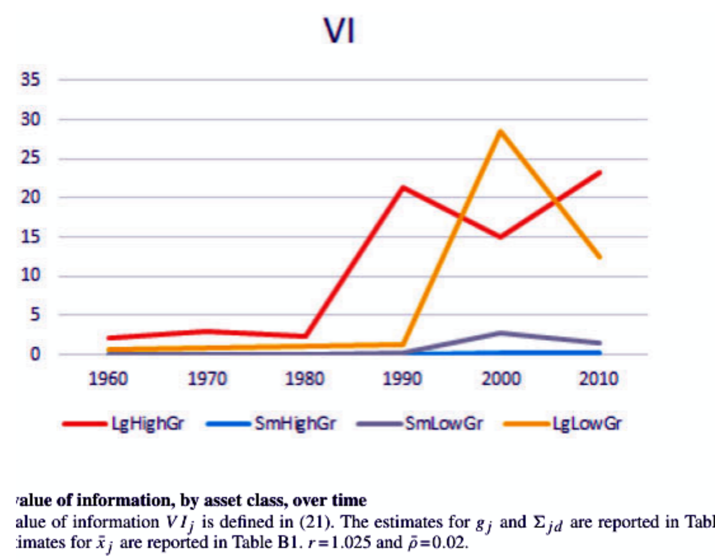


Figure 4: The initial value of information, by asset class, over time. The model-implied value of data rises most for large high-growth firms.

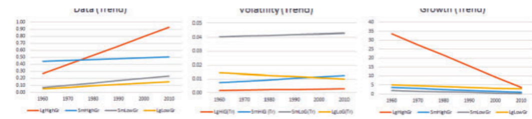
6.5 Figure B1: Time variation in riskless rate; Table B2: Trends in price informativeness and data

Read:

- Figure B1 repeats the component trends allowing the riskless rate to vary over time.
- The qualitative pattern is similar to the baseline: data rises disproportionately for large high-growth firms.
- Table B2 quantifies the trend slopes and reports standard errors.
- The large high-growth data trend is positive and statistically meaningful; the differences relative to other groups are economically large.

Economic message: The baseline result is not driven by assuming a constant riskless rate or by visual inspection alone. Robustness and trend estimates support the data-divergence interpretation.

Interpretation: These appendix visuals strengthen the measurement claim. They show that the main conclusion survives a key input change and that trend differences are not just eyeballed patterns.



igure B1

ime variation in riskless rate

he plots depict a linear trend fitted to the structural estimates of the components of PINF as described in (13) ıd decade-specific interest rates. For details of how the interest rates r are estimated, see the main text.

(a) Figure B1: Time variation in riskless rate.

Table B2

Trends in price informativeness and data

	Trend	SE	Diff in trend	SE
PINF				
Small/high growth	0.0004	(0.0011)	0.0024	(0.0023)
Large/high growth	0.0028	(0.0020)	-	-
Small/ low growth	0.0005	(0.0009)	0.0023	(0.0022)
Large/ low growth	0.0004	(0.0013)	0.0024	(0.0024)
Data				
Small/high growth	0.0207	(0.0800)	0.1072	(0.0873)
Large/high growth	0.1279**	(0.0348)	-	-
Small/ low growth	0.0356	(0.0222)	0.0922**	(0.0413)
Large/ low growth	0.0255	(0.0334)	0.1024**	(0.0482)
Net private data				
Small/high growth	−113.8	(87.6)	1,488.2**	(527.0)
Large/high growth	1,374.4**	(519.7)	-	-
Small/ low growth	15.7**	(5.4)	1,358.8**	(519.7)
Large/ low growth	46.8**	(24.3)	1,327.6**	(520.2)

(b) Table B2: Trends in price informativeness and data.

6.6 Table B1: Number of firms and total assets; Figure D1: Analyst coverage has increased for high-growth firms

Read:

- Table B1 reports the number of firms and average assets by decade and type.
- Large firms, especially large low-growth firms, become much larger over time, while the number of firms in some groups changes substantially.
- Figure D1 shows that analyst coverage rises for high-growth firms, especially within large firms.
- Analyst coverage is not the paper's preferred data measure, but it is a useful external proxy for attention and information production.

Economic message: The size facts help explain why some data become more valuable. The analyst coverage pattern offers outside corroboration that information production shifted toward high-growth firms.

Interpretation: Table B1 is about the scale of assets available to learn about; Figure D1 is about one observable form of information production. They are both consistent with the mechanism that attention and data flow toward assets with higher learning value.

	1960s	1970s	1980s	1990s	2000s	2010s
Number of firms						
Small high growth	1,699	4,739	7,224	9,253	6,444	3,505
Large high growth	1,696	4,229	6,270	7,963	5,662	3,327
Small low growth	1,734	4,664	7,229	9,153	6,382	3,472
Large low growth	1,653	4,040	6,146	7,742	5,534	3,272
Average assets ($ millions)						
Small high growth	125	173	109	175	410	599
Large high growth	2,697	3,510	3,521	8,661	12,928	13,802
Small low growth	517	565	852	2,140	4,478	5,398
Large low growth	6,129	11,592	15,726	22,003	52,550	61,588

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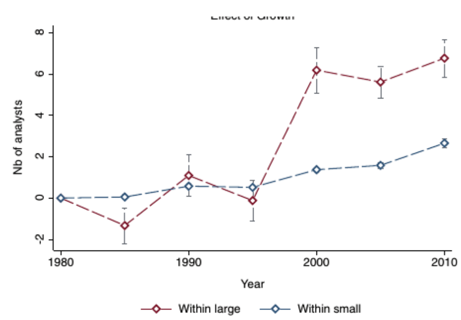
(a) Table B1: Number of firms and total assets by decade and type.



D1

t coverage has increased for high-growth firms

ıph reports coefficients β_t from the following regression: *Number of analysts*$_{it}$ = β_t *Growth*$_i$ × *Half-* + δ_t + e_{it}, where *Growth*$_i$ is a dummy equal to one if the firm is a high-growth firm and *Half-decade*$_t$ nmy for each 5-year interval starting from 1985. We estimate the regression separately for large firms line) and small firms (the blue line).

(b) Figure D1: Analyst coverage has increased for high-growth firms.

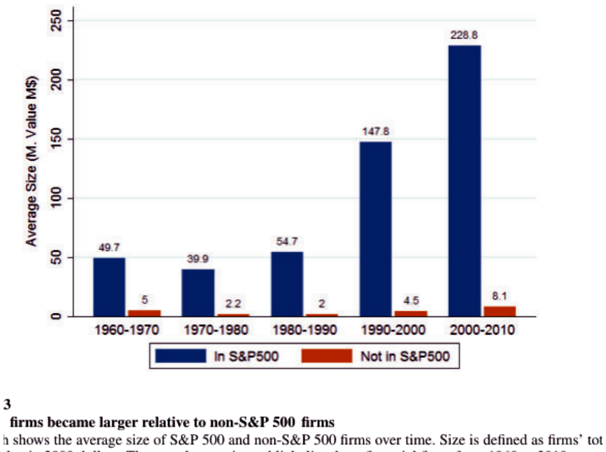
6.7 Figure E3: S&P 500 firms became larger relative to non-S&P 500 firms; Figure E4: Price informativeness by decile

Read:

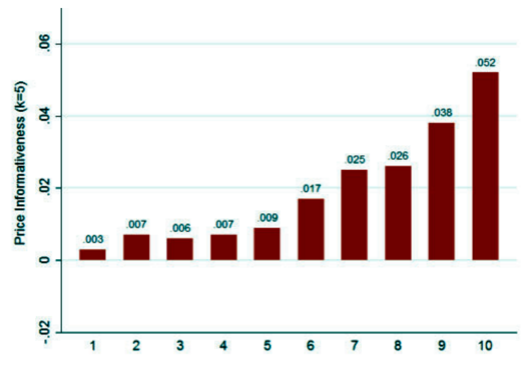
- Figure E3 shows that S&P 500 firms become much larger relative to non-S&P 500 firms over time.
- Figure E4 shows that price informativeness rises with size decile; the highest size deciles are much more informative.
- The evidence supports a size-based interpretation rather than an index-inclusion-only interpretation.

Economic message: If large firms become disproportionately larger, the value of learning about them rises. This helps explain why investors allocate more data processing to the top of the firm-size distribution.

Interpretation: These figures connect the paper's structural mechanism to a simple market fact: firm size concentration. Data concentration and size concentration reinforce each other.



(a) Figure E3: S&P 500 firms became larger relative to non-S&P 500 firms.



(b) Figure E4: Price informativeness by decile.

7 Short summary

- The paper develops a structural measure of investor data precision from prices and cash flows.
- The measure distinguishes data from price informativeness by correcting for growth and volatility.
- The sample uses U.S. CRSP and Compustat data from 1962 to 2016, excluding financial firms.
- Firms are sorted into four groups by size and growth: small/high-growth, large/high-growth, small/low-growth, and large/low-growth.
- The key empirical fact is data divergence: data processing rises sharply for large high-growth firms but not for most other firms.
- The rise for large high-growth firms is driven mainly by net private data, not price information.
- Growth and volatility changes do not mechanically explain the rise; in some cases they work against it.
- The model-implied value of information rises most for large high-growth firms because size and growth make learning more profitable.
- Analyst coverage and reduced-form price informativeness patterns broadly corroborate the structural result.
- The paper suggests that the financial data revolution may improve price discovery mainly for already large, high-growth firms.

8 Conclusion

8.1 Core conclusion

Farboodi, Matray, Veldkamp, and Venkateswaran show that the financial data revolution is uneven. Investors appear to process increasingly abundant data about large high-growth firms, while data about most other firms grows much more slowly. The paper's key contribution is to make this statement quantitative by deriving a structural measure of data precision from price and cash-flow moments.

8.2 Economic intuition

The intuition is that data are valuable only if they can be used profitably. Large firms allow investors to trade larger positions, and high-growth firms have prices that respond more strongly to cash-flow information. Therefore, the same unit of data can be more profitable when applied to a large high-growth firm than to a small or low-growth firm. As the largest firms become even larger, the private return to learning about them rises, and investors have stronger incentives to allocate data-processing resources toward them.

8.3 Why the paper matters

The paper matters because it separates three objects that are often conflated:

- **price informativeness,** a reduced-form predictive relationship between prices and future cash flows;
- **data precision,** the posterior reduction in uncertainty that investors achieve through information processing;
- **value of data,** the private benefit from collecting and using information about a particular asset.

This separation lets the authors show not only that prices of some firms became more informative, but also that investors processed more data about those firms and had economic reasons to do so.

8.4 Policy and market-efficiency implications

The results imply that more data do not automatically make all markets more efficient. If data processing concentrates on firms where information is most profitable, then the largest and most visible firms may become better priced while smaller firms remain informationally neglected. This can matter for capital allocation because financial markets may become increasingly effective at valuing firms that are already large and high-growth, but less effective at disciplining or funding less-followed firms.

8.5 Limitations

The data measure is model-implied, not directly observed. It depends on the noisy rational expectations structure, the removal of common components, the grouping scheme, and the estimated cash-flow and price moments. Alternative models of risk premia, investor heterogeneity, or information markets could imply different mappings from prices to data. The authors address many concerns with robustness and proxy checks, but the results should be read as structural estimates rather than direct observations of data purchases.

8.6 Takeaway

The paper's lasting message is that the data economy in finance is not just growing; it is reallocating attention. Data increasingly flow toward assets where learning is most valuable. This helps explain why information about large high-growth firms has become abundant while information about many other firms has not kept pace.